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Computer Engineering Department



CMPE 310 – Computer Networks and Security

MP7: Skills Integration Challenge

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BSCPE 3-7

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Packet Tracer – Skills Integration Challenge

VLAN Port Assignments and DHCP Information

Ports	VLAN Number - Name	DHCP Pool Name	Network
Fa0/5 – 0/9	VLAN 10 - Sales	VLAN_10	172.31.10.0/ 27
Fa0/10 – Fa0/14	VLAN 20 - Production	VLAN_20	172.31.20.0/ 28
Fa0/15 – Fa0/19	VLAN 30 - Marketing	VLAN_30	172.31.30.0/ 25
Fa0/20 - Fa0/24	VLAN 40 - HR	VLAN_40	172.31.40.0/ 26

Scenario

In this culminating activity, you will configure VLANs, trunks, DHCP Easy IP, DHCP relay agents, and configure a router as a DHCP client.

Requirements

Using the information in the tables above, implement the following requirements:

- Create VLANs on **S2** and assign VLANs to appropriate ports. Names are case-sensitive
- Configure **S2** ports for trunking.
- Configure all non-trunk ports on **S2** as access ports.
- Configure **R1** to route between VLANs. Subinterface names should match the VLAN number.
- Configure **R1** to act as a DHCP server for the VLANs attached to S2.
 - Create a DHCP pool for each VLAN. Names are case-sensitive.
 - Assign the appropriate addresses to each pool.
 - Configure DHCP to provide the default gateway address
 - Configure the DNS server 209.165.201.14 for each pool.
 - Prevent the first 10 addresses from each pool from being distributed to end devices.
- Verify that each PC has an address assigned from the correct DHCP pool.



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Note: DHCP address assignments may take some time. Click **Fast Forward Time** to speed up the process.

- Configure **R1** as a DHCP client so that it receives an IP address from the ISP network.
- Verify all devices can now ping each other and **www.cisco.pka**.



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CONFIGURATION

S1

```
EN
Conf t
vlan 10
vlan 20
vlan 30
vlan 40
interface range fa0/1 - 4, g0/1
switchport mode trunk
Do wr
```

S2

```
enable
config t
!
interface range fa0/1 - 4
switchport mode trunk
!
vlan 10
name Sales
vlan 20
name Production
vlan 30
name Marketing
vlan 40
name HR
!
interface range fa0/5 - 24
switchport mode access
!
interface fa0/5
switchport access vlan 10
```



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```
interface fa0/12
switchport access vlan 20
interface fa0/15
switchport access vlan 30
interface fa0/20
switchport access vlan 40
!
end
wr
```

S3

```
EN
Conf t
vlan 10
vlan 20
vlan 30
vlan 40
interface range fa0/1 - 4
switchport mode trunk
Do wr
```

R1

```
enable
config t
!
ip dhcp excluded-address 172.31.10.1 172.31.10.10
ip dhcp excluded-address 172.31.20.1 172.31.20.10
ip dhcp excluded-address 172.31.30.1 172.31.30.10
ip dhcp excluded-address 172.31.40.1 172.31.40.10
!
ip dhcp pool VLAN_10
network 172.31.10.0 255.255.255.224
default-router 172.31.10.1
```



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```
dns-server 209.165.201.14
ip dhcp pool VLAN_20
network 172.31.20.0 255.255.255.240
default-router 172.31.20.1
dns-server 209.165.201.14
ip dhcp pool VLAN_30
network 172.31.30.0 255.255.255.128
default-router 172.31.30.1
dns-server 209.165.201.14
ip dhcp pool VLAN_40
network 172.31.40.0 255.255.255.192
default-router 172.31.40.1
dns-server 209.165.201.14
!
interface GigabitEthernet0/0
no shutdown
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 172.31.10.1 255.255.255.224
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 172.31.20.1 255.255.255.240
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 172.31.30.1 255.255.255.128
!
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 172.31.40.1 255.255.255.192
!
interface GigabitEthernet0/1
ip address dhcp
no shutdown
!
end
wr
```



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OUTPUTS

Overall Feedback Assessment Items Connectivity Tests

Congratulations! You successfully completed the Packet Tracer - Skills Integration Challenge activity.

Score	: 100/100
Item Count	: 77/77
Component	Items/Total Score
DHCPv4 Client Configuration	24/24 27/27
DHCPv4 Server Configuration	24/24 44/44
Inter-VLAN Routing Configuration	13/13 13/13
Switchport Access Configuration	4/4 4/4
Trunk Configuration	4/4 4/4
VLAN Configuration	8/8 8/8

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP

☒ Static

DHCP request successful.

IPv4 Address

172.31.10.11

Subnet Mask

255.255.255.224

Default Gateway

172.31.10.1

DNS Server

209.165.201.14



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PC2

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static DHCP request successful.

IPv4 Address 172.31.20.11

Subnet Mask 255.255.255.240

Default Gateway 172.31.20.1

DNS Server 209.165.201.14

PC3

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static DHCP request successful.

IPv4 Address 172.31.30.11

Subnet Mask 255.255.255.128

Default Gateway 172.31.30.1

DNS Server 209.165.201.14

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static DHCP request successful.

IPv4 Address 172.31.40.11

Subnet Mask 255.255.255.192

Default Gateway 172.31.40.1

DNS Server 209.165.201.14